

IEEE 34 Nodes Test Feeder Simulation Data for High Impedance Fault Detection and Localization

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1 Abstract

The detection and localization of high impedance faults within distribution power networks is one of the biggest challenges in the field of power grid protection . Over the past two decades, numerous approaches for high-impedance fault detection and localization have been proposed to address this challenge. However, a substantial portion of these works tested their approaches using private real-world or simulation data characterized by diverse power system structures and parameters. This heterogeneity poses a substantial impediment to comprehensive performance evaluation and meaningful comparisons. To fill the blank of benchmark data, we simulate a widely used power system-the IEEE 34-node test feeder- with the introduction of high-impedance fault through Matlab/Simulink. The resulting dataset contains various fault locations distributed across the power grid, two operational disturbances: load switching and capacitor switching, and a normal state with varying operation conditions. Furthermore, the simulation incorporates four distinct high-impedance fault settings commonly utilized in existing studies. The voltage and current signals are collected at all 34 nodes as well as the high-impedance fault module to provide comprehensive system information. Consequently, we hereby make this dataset accessible to researchers engaged in high-impedance fault detection and localization, relieving them from the burden of creating their own datasets and enabling reasonable comparisons of their methodologies with those of their peers.

2 Simulation of IEEE 34-nodes test feeders

Simulink is a powerful tool for simulating a complex power system and generating transient voltage and current signals, which is used to simulate the IEEE 34-nodes test feeder in this work. The IEEE 34-nodes test feeder is an actual feeder operating at a 24.9 kV voltage level in Arizona. As illustrated in Fig.1, the system comprises 34 nodes, including two in-line regulators, an in-line transformer, unbalanced loading, and shunt capacitors.

During simulations, we collected and saved the 3-phase voltage and current signals of each node with the sampling frequency of 30.72 kHz (512 samples each cycle with the 60Hz fundamental frequency) to provide the possibility of studying the effectiveness of node selection for HIF detection. The time duration of each simulation is 0.1 seconds. To simulate actual running conditions of power systems, we considered varying load conditions during each simulation, that is, randomly selecting 50% of nodes and changing their loads proportionally over a range of $[-20\%, +20\%]$. In addition, white noise at 60dB SNR level is added to voltage signals to simulate a real noise environment that is typically in the range of $[48, 70]$ dB SNR [1, 2].

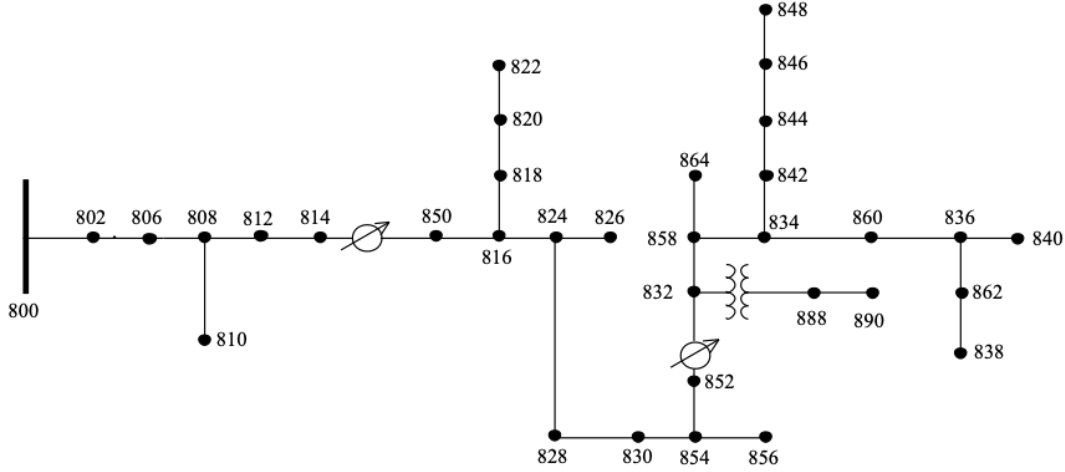


Fig. 1. IEEE 34 Node Test Feeder

3 Simulation of HIF

Fig.2 shows a widely used model that can simulate the dynamic process of high-impedance faults with low complexity and acceptable accuracy [3, 4]. This module's implementation in Simulink consists of two anti-parallel branches, each containing a DC power source, a diode, and a variable resistor connected in series, as illustrated in Fig.3. The two DC sources, V_p and V_n , connected via two diodes, represent the breakdown voltage in different half-cycles, whose magnitudes, therefore, should be unequal and vary randomly to achieve asymmetry and intermittence arcs. The two variable resistors are employed to implement the varying arc resistance, whose value should change randomly in a range with time. When the input voltage of the HIF model V_i is higher than V_p , the left branch is on with current flow from input to output end. Similarly, when $V_i < V_n$, the right branch is on with current flow toward an opposite direction, and the current is close to zero when $V_n < V_i < V_p$. Therefore, this model is characterized by three variables, fault resistance R and two breakdown voltages V_p and V_n , which follow uniform distributions $\mathcal{U}(R_L, R_H)$, $\mathcal{U}(V_{p,L}, V_{p,H})$ and $\mathcal{U}(V_{n,L}, V_{n,H})$, respectively, with a sampling frequency of 1kHz.

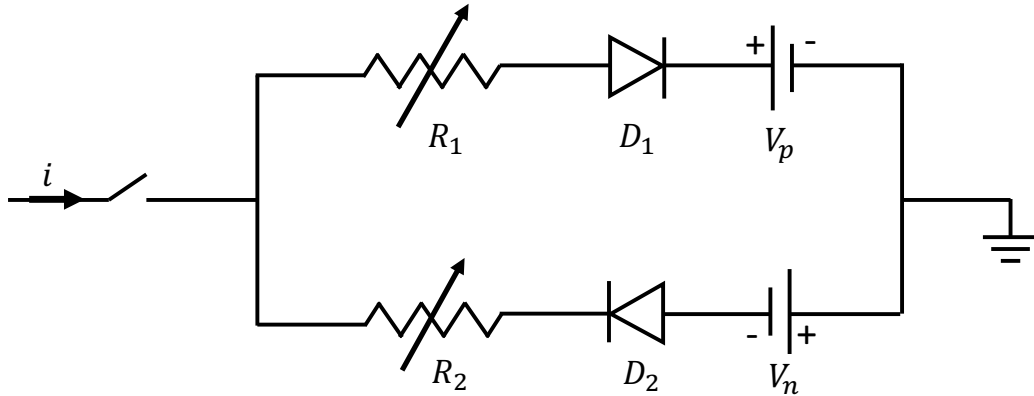


Fig. 2. High impedance fault model

Since the parameters of the HIF model are one of the major factors deciding the difficulty of the detection task, this work considered 4 different HIF conditions, which have been used in the existing

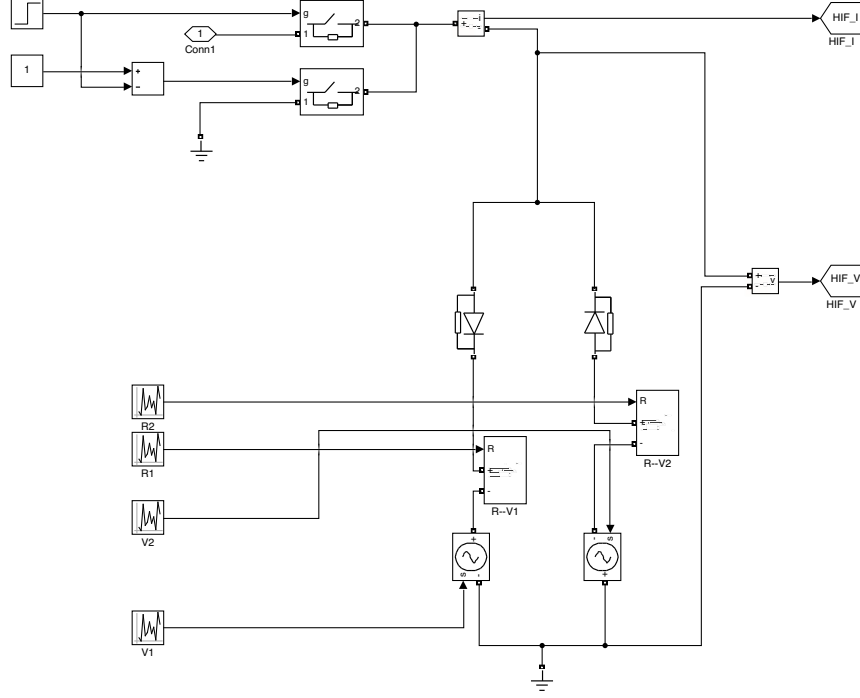


Fig. 3. High impedance fault Simulink module

works [5, 6, 7, 3, 8, 9], for comprehensively evaluating our approach's performance. These parameters are summarized in Tab.1.

When a HIF is introduced in a power system, the HIF module is connected to power delivery line between any two adjacent nodes. For all HIF simulations, faults were always introduced to phase A at 0.03s over 0.1s duration. As demonstrated in Fig.1, this considers 7 different HIF positions near nodes 806, 814, 818, 830, 832, 844, and 860.

Table 1: Parameters of the HIF model

Condition index	$R(\Omega)$	$V_p(kV)$	$V_n(kV)$
1	600 - 1400	5 - 6	7- 8
2	100 - 150	0.9 - 1.1	0.45 - 0.6
3	200 - 1500	5 - 6	7- 8
4	300 - 1000	6.25 - 12.5	6.25 - 12.5

4 Disturbances

The actual operation of a power grid usually contains two primary disturbances, namely load switching and capacitor switching, which may cause false HIF detection results. Referring to works [5, 3], we introduced load switching and capacitor switching events in the system around nodes 890 and 844, respectively, to observe the robustness of the proposed method against disturbances. Load switching events were introduced to all phases at 0.03s with active power varying in the range of [0.45, 0.6]MW. Regarding capacitor switching, a three-phase capacitor bank with random reactive power value in the range of [50, 100]kVAR was switched on at 0.03s.

5 Data collection and format

Table 2: Signals index and annotation

Index	Signal	Index	Signal	Index	Signal	Index	Signal
1	V of 800	2	I between 800 and 802	3	V of 802	4	I between 802and 806
5	V of 806	6	I between 806 and 808	7	V of 808	8	I between 808 and 812
9	V of 810	10	I between 808 and 810	11	V of 812	12	I between 812 and 814
13	V of 814	14	I between 814 and 850	15	V of 816	16	I between 816 and 824
17	V of 818	18	I between 816 and 818	19	V of 820	20	I between 818 and 820
21	V of 822	22	I between 820 and 822	23	V of 824	24	I between 824 and 828
25	V of 826	26	I between 824 and 826	27	V of 828	28	I between 828 and 830
29	V of 830	30	I between 830 and 854	31	V of 854	32	I between 854 and 852
33	V of 856	34	I between 854 and 856	35	V of 832	36	I between 832 and 852
37	V of 858	38	I between 858 and 834	39	V of 864	40	I between 864 and 858
41	V of 888	42	I between 842 and 844	43	V of 890	44	I between 890 and 888
45	V of 834	46	I between 834 and 860	47	V of 842	48	I between 842 and 844
49	V of 844	50	I between 844 and 846	51	V of 846	52	I between 846 and 848
53	V of 848	54	I between 848 and 846	55	V of 860	56	I between 860 and 836
57	V of 836	58	I between 836 and 840	59	V of 840	60	I between 836 and 840
61	V of 862	62	I between 836 and 862	63	V of 838	64	I between 862 and 828
65	V of 850	66	I between 850 and 816	67	V of 852	68	I between 852 and 832
65	V of HIF	66	I enter HIF				

Based on the above description, we generated data using the simulation model, and collected 50 groups of data for each system operation under normal, 2 disturbed, and 28 faulty states (7 positions multiple by 4 HIF conditions), respectively. Additionally, we generated another data set without containing HIF data to enable the training of unsupervised learning model. Each data group records the voltage and current signals of 34 nodes as well as the HIF model during 0.1s (3073 samples). The main parameters of the data set are highlighted in Tab.4 and signals' notation in a data group are given in Tab.2.

Data is saved in .mat format with the following name rules:

Case-[case index]-[condition index]-[realization index].mat

- Case index indicates the system's state. It takes a value from 1 to 10, and their meanings are given in Tab.3.
- Condition index indicates which HIF setting is used to simulate a HIF. When the case index takes a value from 2 to 8, the meaning of the condition index is given in Tab.1. Otherwise, the condition index means nothing but another realization of this case.
- Realization index indicates different repetition of the simulation. It takes a value from 1 to 50.

With the data file loaded, two variables, *Signals* and *t*, will be obtained. Variable *t* is a vector with length *L*, indicating the timestamp of each sampling point. Variable *Signals* is a matrix with length *L* and 66 columns. The meaning of each column is given in Tab.2.

Table 3: Case index notation

Case index	State	Position
1	Normal	-
2	HIF	806
3	HIF	814
4	HIF	818
5	HIF	830
6	HIF	832
7	HIF	844
8	HIF	860
9	Load Switching	890
10	Capacitor switching	844

Table 4: Parameters of the IEEE 34 test feeder data set

Parameter	Value
Power line voltage level (p-p)	24.9kV
Number of nodes	34
Signals sampling frequency	30.72 kHz
Fundamental frequency	60Hz
Simulation time	0.1 sec
Fault occurrence time	0.03 sec
Load switch time	0.03 sec
Capacitor switch time	0.03 sec
Training data groups	50
Testing data groups	1550
Sampling frequency of HIF parameters	1kHz
Load switching range	[0.45, 0.6] MW
Capacitor switching range	[500, 100]kVAR

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